Workshop, Laboratory, Design and Industrial Safety

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Introduction

Industrial safety is the care we take so that no worker is hurt while doing a job. In a workshop, a laboratory, and a factory, machines, electricity, and tools can cause a serious accident if they are used in the wrong way. A small mistake, such as leaving a wire open or the floor wet, can injure a person and also stop the work. Safety is not only a rule on paper. It is a habit of every worker and of the supervisor who watches the workplace. When safety is followed, people stay healthy, machines last longer, and the work is finished without loss.

1. Safety

Safety means staying free from danger, injury, and loss. In an electrical workshop it means that the worker, the tools, and the machines are protected before any work starts. A safe place has covered wires, guarded machines, enough light, and a clear way out. Safety also means the worker knows the job and does not take a risk just to finish quickly. If safety is ignored, one shock or one cut can stop the whole job and harm a life. So safety is the first duty in industry, not a step taken only after an accident.

2. Safety Rules of Industry

These rules are followed so that a small danger does not turn into an accident.

1. Wear a helmet, gloves, goggles, and safety shoes, because a falling part or a spark can injure the head, the eyes, the hands, and the feet.
2. Do not run a machine until you have been trained, because an untrained hand can start it in the wrong way.
3. Switch off the supply and keep it off before any repair, so that no one receives an electric shock.
4. Keep the floor clean and dry, because oil, water, and scrap can make a person slip into a moving machine.
5. Keep the passages and the exits clear, so that workers can leave quickly if there is a fire or another emergency.
6. Report a broken guard, an open wire, or any other danger to the supervisor at once, before another worker is hurt.

3. Causes of Accident

An accident usually happens when an unsafe act and an unsafe condition come together. The worker may be in a hurry, and the machine may already be faulty. If the cause is known, it can be removed before someone is injured.

1. Carelessness, such as talking or looking away while a machine is running.
2. Working without safety equipment, so nothing stands between the body and the hazard.
3. Lack of training, because the worker does not know the safe way to start, stop, or repair the machine.
4. Faulty machines and broken tools, which can slip, spark, or break during the job.
5. Poor light and a slippery floor, which hide the danger and make the worker fall.
6. Not following the safety rules, even after those rules have been written and explained.

4. Safe Acts

A safe act is the correct action a worker takes so that he, and the people near him, stay unhurt.

1. Wearing a helmet, gloves, goggles, and safety shoes before touching a machine or a live circuit.
2. Using the right tool for the work, because a wrong tool can slip and cause a cut or a short circuit.
3. Following the written safety rules and the orders of the supervisor.
4. Keeping the bench and the floor clean, so that tools and wires are not left in the path.
5. Switching the machine off and removing the plug after the work is finished.
6. Telling the supervisor at once if a guard is missing or a wire is left open.

5. Unsafe Acts

An unsafe act is a wrong action by the worker. It is one of the main reasons an accident takes place.

1. Working on a live wire or a machine without a helmet, gloves, or goggles.
2. Starting a machine without training and without knowing how to stop it.
3. Taking the guard off a machine so that the work seems faster.
4. Using a broken tool or a damaged cable, which can fail while it is in the hand.
5. Running, pushing, or playing in the workshop, where machines and hot parts are close.
6. Ignoring a safety rule after it has already been told, which puts the whole group in danger.